MATH 105LA Computer Assignment 1

Due: Monday 10/19, 10pm

1. Use fixed-point iteration to find a fixed point of $g(x) = x/2 + 1/x$ in $[1, 2]$ to approximate $\sqrt{2}$ accurately to within 10^{-5}. Find a root of $f(x) = x - g(x)$ using Newton's method under the same tolerance level. Which method converges faster?

2. An object falling vertically through the air is subjected to viscous resistance as well as to the force of gravity. Assume that an object with mass m dropped from a height s_0 and that the height of the object after t seconds is

$$s(t) = s_0 - \frac{mg}{k}t + \frac{m^2 g}{k^2}(1 - e^{-kt/m}), \tag{1}$$

where $g = 32.17$ ft/s^2 and k represents the coefficient of air resistance in lb-s/ft. Suppose $s_0 = 300$ ft, $m = 0.25$ lb, and $k = 0.1$ lb-s/ft. Find, to within 0.01 second, the time it takes this quarter-pounder to hit the ground.